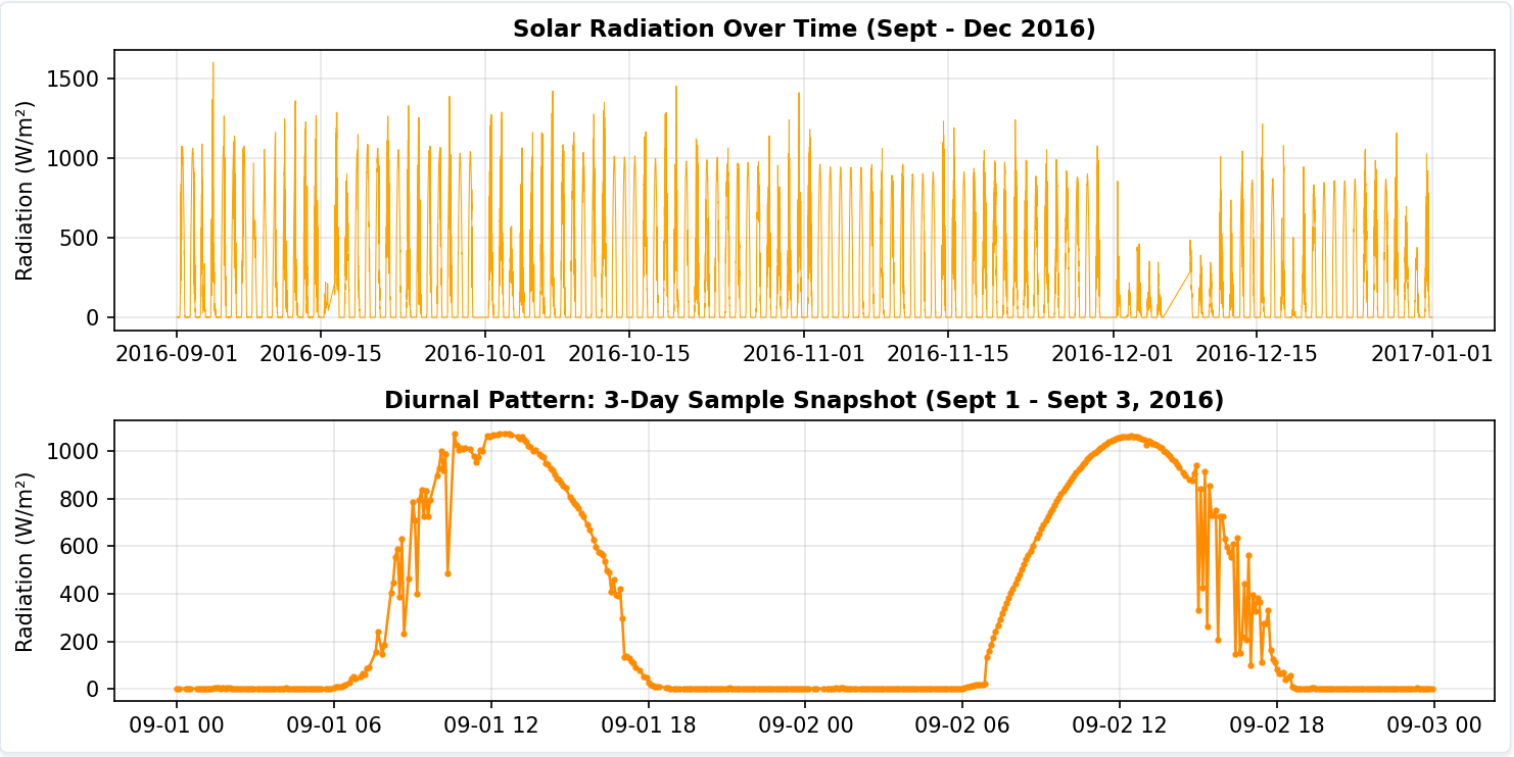
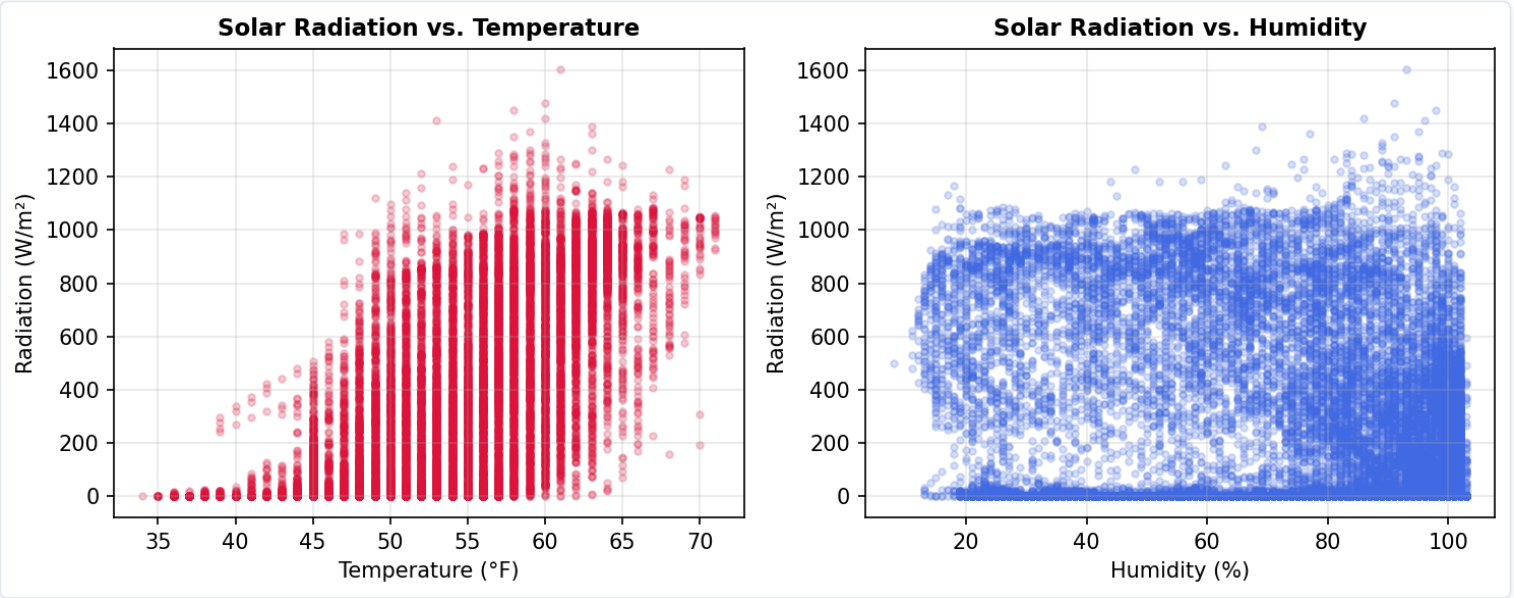
STAC WARS: Solar Radiation Analysis & Prediction Report

Problem Statement 2 Submission Report | Dataset: Voyage-123 Solar Observation Logs (32,686 records)

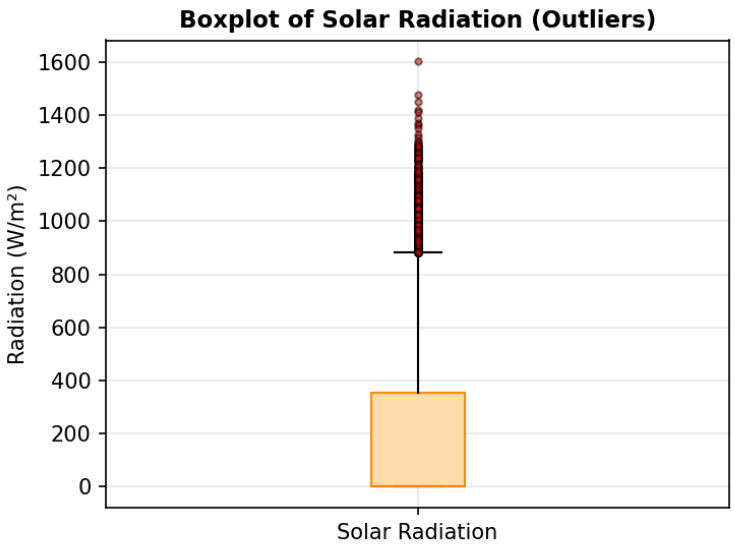
Part A & B: Data Visualization & Key Inferences

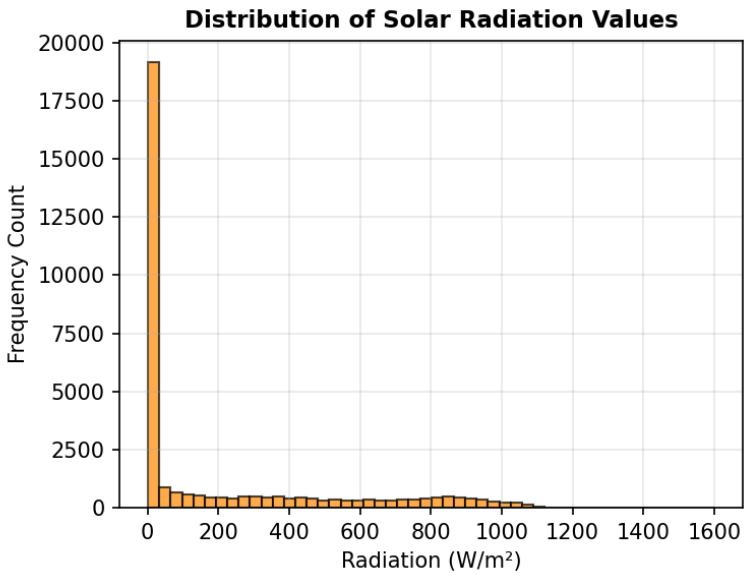
- **Diurnal Peak Timing:** Solar radiation follows a bell-shaped diurnal distribution, peaking between 10:00 AM and 2:00 PM. The maximum average hourly radiation occurs at **Hour 12 (12:00 PM - 1:00 PM)** reaching **727.47 W/m²**.
- **Environmental Parameters:** Temperature shows the strongest positive correlation with solar radiation (**r = +0.7350**). Humidity exhibits an inverse negative relationship (**r = -0.2262**).
- **Outlier Explanation:** Standard atmospheric irradiance is ~1361 W/m². Sensor readings above 1000 W/m² (657 instances) are attributed to **cloud-edge reflection / enhancement effects**.

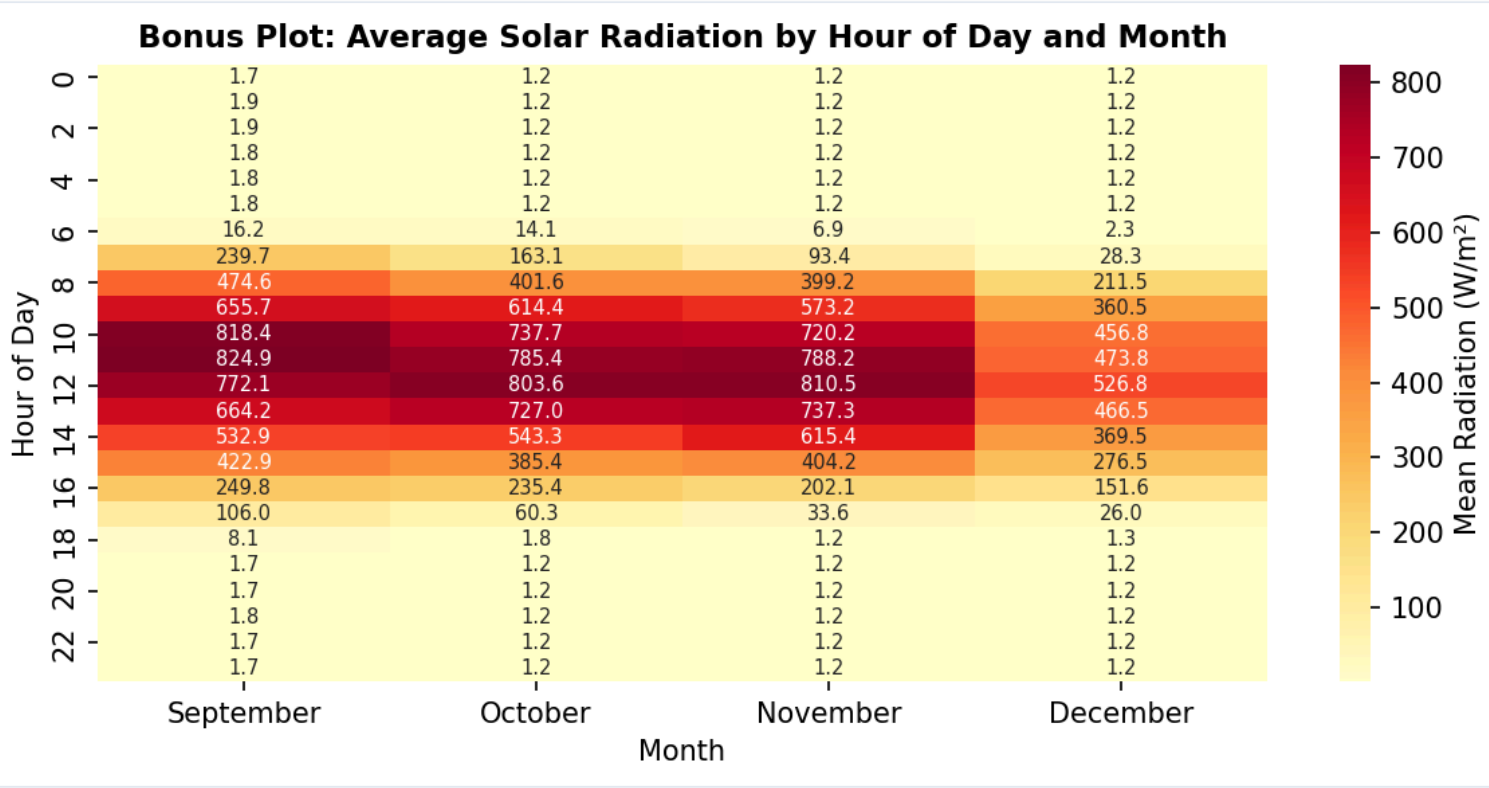




Part A (Cont.): Distribution & Seasonal Diurnal Trends







Part C: Solar Day Activity Analysis

> **Absolute Peak Solar Radiation Event:**
> • **Date & Time:** September 4, 2016 at 12:15:04 PM
> • **Peak Radiation:** 1601.26 W/m²
> • **Environmental Conditions:** Temperature = 61°F | Humidity = 93% | Wind Speed = 9.0 mph
> • **Elapsed Time Since Sunrise: 6 hours, 7 minutes, 4 seconds** after sunrise (Sunrise: 06:08:00 AM)

Top 5 Days with Highest Average Solar Radiation

Rank	Date	Mean Radiation (W/m²)	Max Radiation (W/m²)
1	9/19/2016	335.50	1085.75
2	9/2/2016	333.47	1062.01
3	9/22/2016	325.52	1053.17
4	9/25/2016	319.92	1075.29
5	9/28/2016	312.40	1029.30

Bonus: Machine Learning Model (Solar Radiation Prediction)

- **Model Architecture:** Random Forest Regressor (100 Decision Trees).
- **Target Leakage Avoidance:** Sequential chronological train-test split (80% train / 20% test, `shuffle=False`) to prevent future data from leaking into past training data.
- **Evaluation Metrics: R^2 Score = 0.7301 | RMSE = 137.92 W/m²**
- **Feature Importance:** Temperature is the dominant feature (55.91%), followed by Hour of Day (27.36%).

